

From Classical Theta Functions to the Weil Representation

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Theta Function

Let us start with the Theta Function $\theta : \mathbb{H} \rightarrow \mathbb{C}$,

$$\theta(z) := \sum_{x \in \mathbb{Z}^n} e^{\pi i \|x\|^2 z}$$

θ has some nice transformation rules:

$$\theta\left(\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} z\right) = \theta(z + 2) = \theta(z)$$

$$\theta\left(-\frac{1}{z}\right) = \left(\frac{z}{i}\right)^{\frac{n}{2}} \theta(z) \text{ (This comes from Poisson Summation)}$$

It turns out that θ is a modular form of weight $n/2$ for $\Gamma_0(4)$ if n is even.

Road to The Weil Representation

θ is a summation of the Schwartz Function

$$\varphi_z(x) = e^{\pi i \|x\|^2 z} \in \mathcal{S}(\mathbb{R}^n)$$

over the lattice $\mathbb{Z}^m$

$O(n)$ has a obvious action over $\mathcal{S}(\mathbb{R}^n)$ given by

$$\omega(g)\varphi(x) = \varphi(g^{-1}x).$$

But $SL_2(\mathbb{R})$ also has an interesting action over $\mathcal{S}(\mathbb{R}^n)$ that we can study.

Road to The Weil Representation

Translations $n(b) = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$:

$$n(b)\varphi_z(x) = \varphi_{z+b}(x) = e^{\pi i \|x\|^2 (z+b)} = e^{\pi i b \|x\|^2} \varphi_z(x)$$

Dilations $m(a) = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}$:

$$\begin{aligned} m(a)\varphi_z(x) &= \varphi_{a^2 z}(x) = e^{\pi i \|x\|^2 a^2 z} \\ &= e^{\pi i \|ax\|^2 z} = \varphi_z(ax) \end{aligned}$$

Inversion $w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$:

$$w\varphi_z(x) = \varphi_{-\frac{1}{z}}(x) = e^{-i\pi \|x\|^2 z^{-1}} = \left(\frac{z}{i}\right)^{\frac{n}{2}} \hat{\varphi}_z(x)$$

The Weil Representation

While these aren't exactly the actions that we care about, they are very close. The Weil Representation ω of $\mathrm{SL}_2(\mathbb{R})$ on $\mathcal{S}(\mathbb{R}^n)$ are defined as follows:

Translations $n(b)$:

$$\omega(n(b))\varphi(x) = e^{\pi i b \|x\|^2} \varphi(x)$$

Dilations $m(a)$:

$$\omega(m(a))\varphi(x) = |a|^{\frac{n}{2}} \varphi(ax)$$

Inversion w :

$$\omega(w)\varphi(x) = i^{\frac{n}{2}} \hat{\varphi}(x)$$

Relating the Actions

Both $O(n)$ and $SL_2(\mathbb{R})$ are acting on $\mathcal{S}(\mathbb{R}^n)$. It actually turns out that the actions commute. The only action that technically we need to check is the action of $w \in SL_2(\mathbb{R})$ and $g \in O(n)$. But the good news is that the fourier transform commutes with rotations and reflections!

The Symplectic Group and Dual Pairs

Yes! $O(n), \mathrm{SL}_2(\mathbb{R}) \subset \mathrm{Sp}(2n, \mathbb{R})$ where $\mathrm{Sp}(2n, \mathbb{R})$ and the Weil Representation is actually a Representation of $\mathrm{Sp}(2n, \mathbb{R})$. Furthermore to explain the commutation of the actions, we find out that

$$Z_{\mathrm{Sp}(2n, \mathbb{R})}(O(n)) = \mathrm{SL}_2(\mathbb{R})$$

$$Z_{\mathrm{Sp}(2n, \mathbb{R})}(\mathrm{SL}_2(\mathbb{R})) = O(n)$$

Thus $(\mathrm{SL}_2(\mathbb{R}), O(n))$ form something called a **Reductive Dual Pair**.

Theta Kernel

For our dual pair $(\mathrm{SL}_2(\mathbb{R}), O(n))$ we can we can define a kernel function

$$\Theta_{\Lambda, \varphi} : \mathrm{SL}_2(\mathbb{R}) \times O(n) \rightarrow \mathbb{C}$$

for a Schwartz function φ and a lattice Λ where

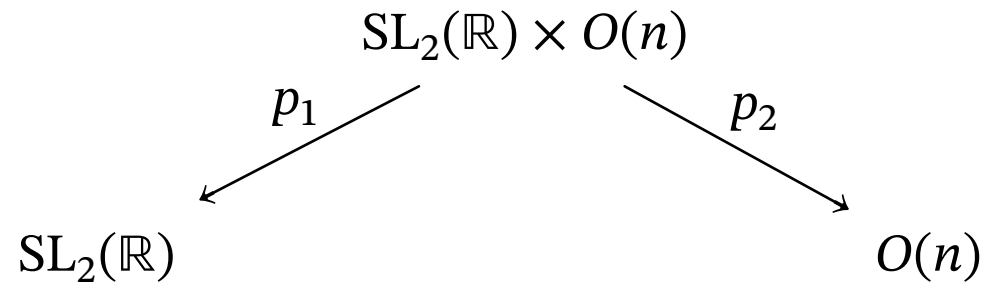
$$\Theta_{\Lambda, \varphi}(g, h) = \sum_{\lambda \in \Lambda} (\omega(g, h)\varphi)(\lambda)$$

Key Property: $\Theta_{\Lambda, \varphi}$ transforms nicely in both variables.

1. In g : It is a automorphic form for $\mathrm{SL}_2(\mathbb{R})$ with respect to some discrete subgroup Γ
2. In h : It is a automorphic form for $O(n)$ with respect to $O(n, \mathbb{Z})$

Theta Lift

Thus $\Theta_{\Lambda, \varphi} : \mathrm{SL}_2(\mathbb{R}) \times O(n) \rightarrow \mathbb{C}$.



So we can get an integral lift by doing the standard pull back, multiply, and push forward method to get an automorphic form from one side to the other. In fact this is how the Theta Series can be constructed representation theoretically.

Howe Duality

Howe Duality: For a dual pair (G, H) , the theta lift is one-to-one (under certain conditions). Thus the representations spaces of $SL_2(\mathbb{R})$ and $O(n)$ paired by the correspondence are unique.

Other Reductive Pairs

1. Symplectic-Orthogonal (The Classic Case): $(O(V), \mathrm{Sp}(W))$. This gives us Shimura correspondence
2. General Linear (GL-GL): $(\mathrm{GL}(n), \mathrm{GL}(m))$ which sit inside Sp_{2mn} . This gives us a convolution formula for L-functions
3. Unitary-Unitary- Useful for studying automorphic forms on unitary groups